

Jira Activity Analyzer v0.2.41 Implementation Report

Result

Implementation completed; external acceptance pending.

實作完成；待外部實機驗收。

v0.2.41 implements Stable Hash V3, evidence-backed Coverage, Current Observed Metrics, Event Identity V2, a new writable database generation, Stage 5 diagnostics, focused automated validation, app-only UI validation, and Windows packaging. No real Jira connection or long packaged smoke was executed.

v0.2.40 Defect Closure

Confirmed defect	v0.2.41 fix	Automated evidence	Real-machine evidence
Volatile-only changes created Updated	Exact verified Field IDs and lastViewed are excluded by frozen Stable Hash V3 policy	Volatile-only fixture remains Existing; revision and payload timestamp unchanged	NOT EXECUTED
Parenthesized Working Days names did not resolve	NFKC normalization handles ASCII/full-width parentheses and spacing, then resolves one exact Jira Field ID	Full-width/spacing mapping test passed	NOT EXECUTED
Policy metadata differed from execution	One canonical policy JSON and fingerprint drive hashing, metadata, diagnostics, and compatibility checks	Determinism and mismatch tests passed	NOT EXECUTED
Issue Links were saved but Coverage said Skipped	Evidence is derived from validated issue.fields.issueLinks data independently of Related Issues	empty/nonempty/missing/count mismatch tests passed	NOT EXECUTED
Edited Comment created another Created event	Created identity is comment_created:<commentId>; Updated adds normalized Jira timestamp	edit/re-fetch tests passed	NOT EXECUTED
Current links generated repeated events	Link events require Jira changelog History ID, item index, and field identity	current-links-only creates zero link events	NOT EXECUTED
Excluded volatile values had no current store	Added current_observed_metrics, one row per Issue and Field ID	insert/update/null/missing/rollback /bounded-row tests passed	NOT EXECUTED

Implemented Modules

- electron/stableIssueContentV3.ts: canonical policy, display-name normalization, exact Field ID resolution, policy fingerprint, Stable Projection V3, SHA-256, safe diff paths, and Current Observed Metric extraction.
- electron/coverageProfile.ts: evidence records with status, count, source, request completion, validation, reason code, Issue Links invariants, and scope comparison.
- electron/activityEvents.ts: Event Identity V2 using Jira-native IDs, deterministic changelog item identity, insert-only dedup candidates, and safe missing-ID warnings.
- electron/currentStateArchive.ts: schema v2, policy freeze, V2/V1 read-only classification, payload prevalidation, metrics, policy enforcement, Coverage-only upgrade, and atomic per-Issue rollback.

- electron/sourceArchiveDatabaseWrite.ts: file-backed Coverage evidence, independent Related Issues semantics, V3 diagnostics, and issue-link count conservation.
- src/routes/AnalysisPage.tsx: bilingual V3/V2 policy, fingerprint, metrics, Coverage, payload, event, and safe diff diagnostics.
- electron/v0241StableHashCoverageEventIdentity.test.ts: focused correctness, integrity, read-only, bounded-storage, and rollback coverage.

Database and Policy

New v0.2.41 databases declare:

- Schema version: 2
- Storage model: current_state
- Storage model version: 1
- Stable Hash policy version: 3
- Event Identity policy version: 2

database_metadata stores canonical policy JSON and SHA-256 fingerprints. Stable policy resolution is frozen atomically on the first eligible save. The Event Identity V2 fingerprint is deterministic:

708ce7c3789678f4580952427b29b7d94742af72e2834e53d7173db8856a5c2a.

The Stable policy fingerprint depends on the authoritative Jira names and schema metadata for that database. Resolved fields store Field ID, Jira display name, configured name, normalized name, and schema summary. Unresolved or ambiguous candidates remain included in Stable Hash and produce warnings.

Current-State row invariants:

- One Source Object, Snapshot, compressed Payload, and Sync State per Issue.
- One Current Observed Metric per Issue and resolved Field ID.
- Multiple Activity Events, unique by (source_object_id, event_identity_hash).
- No Snapshot, Payload, Metric, or Event history was added.

v0.2.39 and v0.2.40 databases are detected read-only. The focused test compares an old-policy file SHA-256 before and after inspection and proves byte identity.

State Machine and Transactions

- New: inserts current rows, metrics, and new events; initializes revision 1.
- Existing: increments successful checks, updates metrics, deduplicates events, and retains Snapshot/Payload bytes and timestamps.
- Meaningful Updated: replaces Snapshot/Payload and increments revision exactly once.
- Coverage Upgrade: replaces the authoritative archive but does not increment revision or update business-content time.
- Partial, Failed, invalid/downgrade/incomparable Coverage, server mismatch, and policy mismatch retain prior formal state.
- Snapshot, Payload, Sync State, Metrics, and Events share one per-Issue SQLite transaction. Failure injection proves full rollback.

Validation

Command	Result	Duration	Notes
npm.cmd run test:v0.2.41	Passed	1.9 s final	Stable Hash V3, Coverage, Metrics, Event Identity V2, DB and rollback
npm.cmd run typecheck	Passed	7.9 s final	Renderer and Electron TypeScript
npm.cmd run test:unit	Passed	28.6 s final	Active aggregate plus v0.2.41 focused suite
npm.cmd run build	Passed	30.6 s	Vite and Electron bundles
npm.cmd run test:ui	Passed	79.1 s final	9 routes x 5 viewports, app-only Electron validation
npm.cmd run dist	Passed	78.6 s	Installer and Portable generated
git diff --check	Passed	under 1 s	No whitespace errors
Changed-file secret scan	Passed	under 1 s	No credential pattern found

Recorded failed/retried validation:

- First v0.2.41 focused run failed because Issue Links validation still accepted a top-level fallback; the fallback was removed and the final run passed.
- First aggregate run reached the historical v0.2.39 event-count expectation. That suite expects synthetic events from current links, which Event Identity V2 intentionally forbids. v0.2.39/v0.2.40 policy-specific commands remain available but are not part of the active v0.2.41 aggregate.
- First UI run failed because the synthetic packaged fixture lacked V3 Coverage evidence. The fixture was upgraded; the final app-only UI run passed.

Known non-blocking warnings:

- Vite reports the existing renderer chunk above 500 kB.
- Electron test mode reports the existing development Content Security Policy warning.
- electron-builder uses the default Electron icon.
- npm reports that a newer major npm version is available.

Artifacts

- Installer: release/Jira Activity Analyzer Setup 0.2.41.exe, 106,475,208 bytes, File/Product version 0.2.41.
- Portable: release/Jira Activity Analyzer Portable 0.2.41.exe, 106,245,134 bytes, File/Product version 0.2.41.
- Implementation report: this Markdown file and the corresponding PDF.
- Manual checklist: reports/JiraActivityAnalyzer_v0.2.41_manual_acceptance_checklist.md.
- Execution ledger: reports/JiraActivityAnalyzer_v0.2.41_execution_time_ledger.json.

Release outputs are ignored and are not committed.

Ten-Scenario Acceptance

Round	Status
1 New database, four real Issues	NOT EXECUTED / 待實機驗收
2 Immediate unchanged refetch	NOT EXECUTED / 待實機驗收
3 View/open only	NOT EXECUTED / 待實機驗收
4 Volatile recalculation only	Automated fixture passed; real Jira NOT EXECUTED
5 One business field edit	Automated fixture passed; real Jira NOT EXECUTED
6 Add one Comment	Automated fixture passed; real Jira NOT EXECUTED
7 Edit same Comment	Automated fixture passed; real Jira NOT EXECUTED
8 Unchanged event refetch	Automated fixture passed; real Jira NOT EXECUTED
9 Invalid Issue Links evidence	PASSED by deterministic test
10 Transaction failure injection	PASSED by deterministic test

Security and Scope

No Jira write method, database migration, old-event cleanup, history model, credential store, database registry, or broad Full Fetch redesign was added. Automated tests use synthetic data. Changed-file scanning found no token, Authorization credential, password, private key, raw Jira dataset, SQLite database, Debug Folder, installer, or Portable artifact staged for Git.

Known Limitations

- Real Jira and manual Windows GUI acceptance were not executed.
- Groovy formula definitions are not visible; only returned values and Jira field metadata can be observed.
- A database created without complete Jira names metadata will retain unresolved fields in Stable Hash by fail-safe design.
- Existing v0.2.39/v0.2.40 semantic duplicate events are not repaired or deleted.
- The long packaged smoke test was intentionally not executed.

Git branch, final commits, push result, and reconciled execution time are finalized in the ledger and final handoff after delivery commits.